
CS 109ACE Week 3

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Agenda + Announcements

Agenda

- Check-in: how did PSET 1 go?
- Review + practice: counting (Lecture 4)
- Review + practice: random variables, PMFs, the Binomial (Lecture 5)
- Review + practice: moments (Lecture 6)
- Challenge problems + your questions

Announcements

- PSET 2 is out: see the course site for the due date
- Midterm: Thu Oct 22, 6:30 pm. That's 2.5 weeks away!

Also...

Please complete the form from last week!

Exit tickets / Google forms will be used to improve the course and verify your attendance

Based on your feedback

- Lecturing was too fast
 - I will remove slides
- I will do more practice
 - And more visual learning

I will continue weekly recap sheets!



My slides will include more questions than we can do in a section. I will pick the best!

Recap Problem

On a baseball team:

- 20% of athletes do not practice

- 60% of athletes practice a lot

- 20% of athletes practice a little

An athlete with a lot of practice is twice as likely to start than an athlete with a little.

An athlete with a little practice is twice as likely to start than an athlete without practice.

Given an athlete started, what is the probability that they practiced a lot?

Recap Problem: Solution

Events: N = no practice, L = a little, A = a lot; S = the athlete started

Priors: $P(N) = 0.2$, $P(L) = 0.2$, $P(A) = 0.6$

Let $P(S | N) = p$, so $P(S | L) = 2p$ and $P(S | A) = 4p$ (p will cancel)

Bayes' rule, with the law of total probability in the denominator:

$$\begin{aligned} P(A | S) &= \frac{P(S | A) P(A)}{P(S | N) P(N) + P(S | L) P(L) + P(S | A) P(A)} \\ &= \frac{4p \cdot 0.6}{p \cdot 0.2 + 2p \cdot 0.2 + 4p \cdot 0.6} = \frac{2.4p}{3.0p} = 0.8 \end{aligned}$$

Answer: $P(A | S) = 0.8$

Independence & Inclusion–Exclusion

Review: The Two Easy Cases

Mutually exclusive ($E \cap F = \emptyset$) makes OR easy:

$$P(E \cup F) = P(E) + P(F)$$

Independent makes AND easy:

$$P(E \cap F) = P(E) \cdot P(F) \quad (\text{equivalently } P(E | F) = P(E))$$

Mutually exclusive events (with positive probability) are NEVER independent

Key idea: disjoint $\rightarrow$ add; independent $\rightarrow$ multiply; else inclusion–exclusion / chain rule

Practice 1: Three Clubs

At a school, 50% of students are in a CS club, 40% in a math club, and 30% in a stats club. 20% are in CS and math, 15% in CS and stats, 10% in math and stats, and 5% are in all three.

- (a) What fraction of students are in at least one of the clubs?
- (b) What fraction are in none?
- (c) What fraction are in the CS club only?

Practice 1: Solution

Let C, M, T = in the CS, math, stats club.

(a) Inclusion–exclusion:

$$P(C \cup M \cup T) = 0.50 + 0.40 + 0.30 - 0.20 - 0.15 - 0.10 + 0.05 = 0.80$$

(b) Complement: $P(\text{none}) = 1 - 0.80 = 0.20$

(c) CS only = $P(C) - P(CM) - P(CT) + P(CMT)$ (we subtracted the triple overlap twice)
 $= 0.50 - 0.20 - 0.15 + 0.05 = 0.20$

Check: draw out Venn regions!

Answers: (a) 0.80 (b) 0.20 (c) 0.20



Counting



Warm-up: Entry Codes

Entry codes are 6 digits and may start with 0 (e.g., 379009, 012345, 000000).

- (a) How many codes are possible?
- (b) How many are the same two digits repeated three times (121212, 333333)?
- (c) How many use each of the digits 1–6 exactly once (254631)?
- (d) Stretch: a code is valid if its digits sum to exactly 9. What is $P(\text{valid})$ for a code picked uniformly at random?

Warm-up: Solution (1/2)

(a) 10 choices for each of 6 positions (product rule): $10^6 = 1,000,000$

(b) The first two digits fix the whole code: $10 \cdot 10 = 100$

(c) Arrange 6 distinct digits in a row: $6! = 720$

(Careful in (b): 333333 counts once, as the pair “33”.)

Warm-up: Solution (2/2)

(d) Count digit strings with $d_1 + d_2 + \dots + d_6 = 9$, where each digit ≥ 0

No digit can exceed 9, since the total is only 9

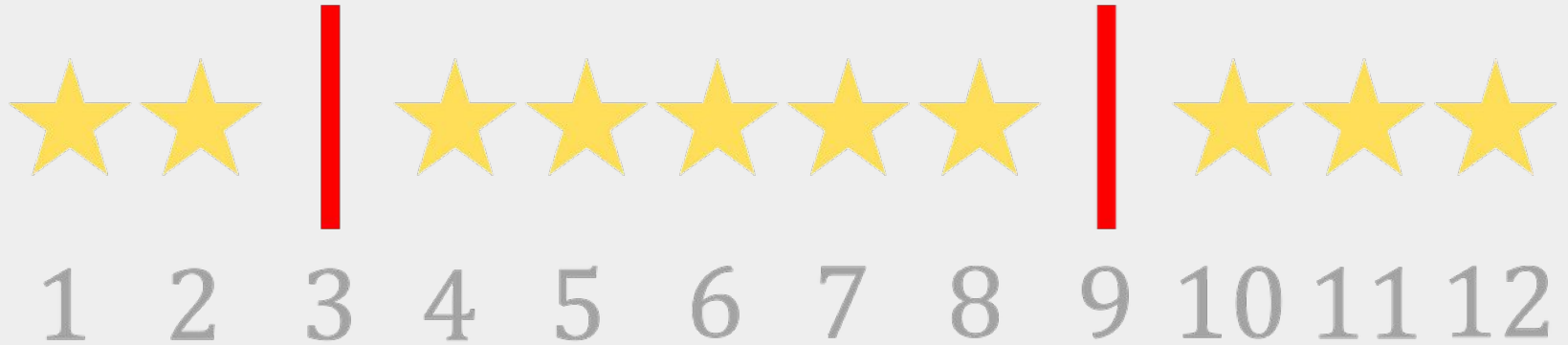
Stars and bars: 9 stars and 5 bars in a row $\rightarrow C(9+6-1, 6-1) = 2002$

$P(\text{valid}) = 2002 / 10^6 \approx 0.002$

Answers: (a) 10^6 (b) 100 (c) 720 (d) $2002 / 10^6 \approx 0.002$

Extra: Stars and Bars

10 identical units among 3 labeled bins



$(10 + (3-1))$ choose $(3-1)$

Review: Counting Rules

Step (product) rule: m ways for step 1, n ways for step 2 (each time) $\rightarrow m \cdot n$ total

Does order matter? Are the objects distinct?

	Distinct objects	Some identical
Sort all n	$n!$	$n! / (n_1! n_2! \cdots n_r!)$
Choose k , unordered	$C(n, k) = n! / (k! (n - k)!)$	stars and bars
Choose k , ordered	$n! / (n - k)!$	—

Permutations with Repeats

BAYES: 5 distinct letters $\rightarrow 5! = 120$ orderings

BOBA: $4! = 24$ if the B's were different, but every word is counted $2!$ times
 $\rightarrow 4! / 2! = 12$

MISSISSIPPI: 11 letters (1 M, 4 I, 4 S, 2 P) $\rightarrow 11! / (1! 4! 4! 2!) = 34,650$

Why divide? Label the copies (B_1, B_2), count all $n!$ orderings, then divide by the number of ways to shuffle each group of identical copies

Key idea: n semi-distinct objects $\rightarrow n! / (n_1! n_2! \cdots n_r!)$ orderings

Combinations

20 people, choose 5 to get cake (order doesn't matter):

1. Line up all 20 people: $20!$ ways
2. First 5 in line get cake
3. Every cake group was counted $5! \cdot 15!$ times (reorder front 5, back 15)
→ $20! / (5! 15!) = C(20, 5) = 15,504$

Symmetry: $C(n, k) = C(n, n - k)$ (choosing who gets cake = choosing who doesn't)

Challenge: can someone prove this? Can someone explain why true?

Key idea: $C(n, k) = n! / (k! (n - k)!)$ counts unordered groups of k from n distinct objects

Practice 2: Committees

- (a) An animal club has 9 antelopes and 11 beavers. How many 7-member committees have exactly 4 beavers and 3 antelopes?
- (b) From 2 cats and 5 dogs, a committee of 3 is formed. How many committees include at least one cat?

Practice 2: Solution

(a) Two independent steps (product rule):

choose 4 of 11 beavers, 3 of 9 antelopes: $C(11, 4) \cdot C(9, 3) = 330 \cdot 84 = 27,720$

(b) “At least one” $\rightarrow$ complement: all committees minus cat-free committees

$$C(7, 3) - C(5, 3) = 35 - 10 = 25$$

Answers: (a) 27,720 (b) 25

Practice 3: BASKETBALL

- (a) How many distinct 10-letter arrangements of the letters in BASKETBALL are there?
- (b) How many of those arrangements have the two L's next to each other?

Practice 3: Solution

(a) BASKETBALL has 10 letters: B $\times 2$, A $\times 2$, L $\times 2$, and S, K, E, T once each

$$10! / (2! \cdot 2! \cdot 2!) = 3,628,800 / 8 = 453,600$$

(b) Glue the two L's into one block “LL”: now there are 9 objects (B $\times 2$, A $\times 2$, LL, S, K, E, T)

$$9! / (2! \cdot 2!) = 362,880 / 4 = 90,720$$

(no extra factor for the block: the two L's are identical)

Answers: (a) 453,600 (b) 90,720

Practice 4: Bookshelf

3 different hardcover books and 5 different paperbacks go on a shelf.

How many arrangements keep all the hardcovers together AND all the paperbacks together?

Practice 4: Solution

Steps (product rule):

1. Order the two blocks (hardcovers first or paperbacks first): $2! = 2$
 2. Order the hardcovers inside their block: $3! = 6$
 3. Order the paperbacks inside their block: $5! = 120$
- $$2 \cdot 6 \cdot 120 = 1440$$

(Out of $8! = 40,320$ total arrangements, only 1440 keep both groups together)

Answer: 1440

Refresher: Stars and Bars

Put n IDENTICAL items into r DISTINCT bins:

$$\# \text{ ways} = C(n + r - 1, r - 1)$$

“Each bin gets at least one”: hand one item to each bin first, then split the remaining $n - r$

$$\# \text{ ways} = C(n - 1, r - 1)$$

Key idea: n identical items into r bins $\rightarrow C(n + r - 1, r - 1)$

Practice 5: Splitting \$7

You have 7 identical \$1 bills to split among Amber, Ben, and Curtis.

- (a) How many ways can you split them (someone may get \$0)?
- (b) How many ways if each person gets at least \$1?

Practice 5: Solution

(a) 7 stars, 2 bars (3 people $\rightarrow$ 2 dividers): $C(7 + 2, 2) = C(9, 2) = 36$

(b) Give everyone \$1 first, then split the remaining \$4 freely:

$$C(4 + 2, 2) = C(6, 2) = 15$$

Example: ★★ | ★ | ★★★★★ in (a) means Amber \$2, Ben \$1, Curtis \$4

Answers: (a) 36 (b) 15

Counting Challenge: Six-Card Poker

A casino deals 6-card poker hands from a standard 52-card deck (hands are unordered).

(a) Three pair: 2 cards of one rank, 2 of another, and 2 of a third. What is $P(\text{three pair})$?

(b) Pyramid: 3 cards of one rank, 2 of another, and 1 of a third. What is $P(\text{pyramid})$?

Solution: Six-Card Poker

Sample Space: $|S| = C(52, 6) = 20,358,520$ unordered hands

(a) Three pair: Choose 3 ranks, then choose 2 suits for each rank:

$$|E| = C(13, 3) \cdot C(4, 2)^3 = 286 \cdot 216 = 61,776 \rightarrow P(\text{three pair}) = 61,776 / 20,358,520 \approx \mathbf{0.3034\%}$$

(b) Pyramid (3–2–1): Choose distinct ranks for triple, pair, single ($13 \cdot 12 \cdot 11$), then pick suits:

$$|E| = (13)(12)(11) \cdot C(4, 3) \cdot C(4, 2) \cdot C(4, 1) = 1,716 \cdot 96 = 164,736 \rightarrow P(\text{pyramid}) = 164,736 / 20,358,520 \approx \mathbf{0.8092\%}$$

Answer: $P(\text{three pair}) \approx 0.3034\%$, $P(\text{pyramid}) \approx 0.8092\%$

Poker Straight

Draw a 5-card hand from a standard 52-card deck.

A straight is 5 cards of consecutive ranks, e.g., 2-3-4-5-6. An ace can be low (A-2-3-4-5) or high (10-J-Q-K-A). Suits don't matter (count straight flushes too).

What is $P(\text{straight})$?

First, describe the sample space and the event in words.

Poker Straight: Solution

S = all 5-card hands (unordered): $|S| = C(52, 5) = 2,598,960$

E = all straights. Build one step by step (product rule):

Pick the lowest rank (A, 2, ..., 10): 10 ways, which fixes all 5 ranks

Pick a suit for each card: $4^5 = 1024$ ways

$$|E| = 10 \cdot 1024 = 10,240 \rightarrow P(\text{straight}) = 10,240 / 2,598,960 \approx 0.0039$$

Why not $\times 2$ for “up or down”? Naming a straight by its lowest card counts each hand exactly once.

Answer: $P(\text{straight}) = 10,240 / 2,598,960 \approx 0.0039$

Random Variables

Random Variables

A random variable (RV) is a number whose value is determined by a random outcome

X = result of one die roll; Z = sum of two dice; Y = # heads in n flips

An RV is NOT an event, but “ $X = k$ ” IS an event, so $P(X = k)$ makes sense

$P(Z = 2) = 1/36$, $P(Z = 7) = 6/36$, $P(Z = 13) = 0$

Writing $P(X)$ alone means nothing; always say which value. Sometimes abbreviated as $P(x)$

Key idea: random variable = number from a random outcome; $X = k$ is an event

Probability Mass Functions (PMFs)

The PMF of a discrete RV X lists $P(X = k)$ for every value k

Can be a table, a formula, a graph, or code. Two dice ($Z = \text{sum}$):

$$P(Z = k) = \begin{cases} (6 - |k - 7|) / 36 & \text{if } k \in \{2, 3, \dots, 12\} \\ 0 & \text{otherwise} \end{cases}$$

Rules every PMF must satisfy: $0 \leq P(X = k) \leq 1$ and $\sum P(X = k) = 1$ (sum over all k)

PMFs others hand you work too: $P(X = k) = 1/2^k$ for $k = 1, 2, \dots$ (0 otherwise)

Key idea: a PMF maps each value k to $P(X = k)$, and the values must sum to 1

RV Review Through Example: The CDF

Same X (one fair die). The CDF “looks up” the probability that X is at or below a :

$$F(a) = P(X \leq a) = \sum p(x) \text{ over all } x \leq a$$

$$F(a) = \begin{cases} 0 & \text{if } a < 1 \\ 1/6 & \text{if } 1 \leq a < 2 \\ 2/6 & \text{if } 2 \leq a < 3 \\ 3/6 & \text{if } 3 \leq a < 4 \\ 4/6 & \text{if } 4 \leq a < 5 \\ 5/6 & \text{if } 5 \leq a < 6 \\ 1 & \text{if } a \geq 6 \end{cases}$$

$$\begin{aligned} \text{So } F(2) &= P(X \leq 2) = P(X \text{ takes on the value 1 or 2}) \\ &= p(1) + p(2) = 2/6 = 1/3 \end{aligned}$$

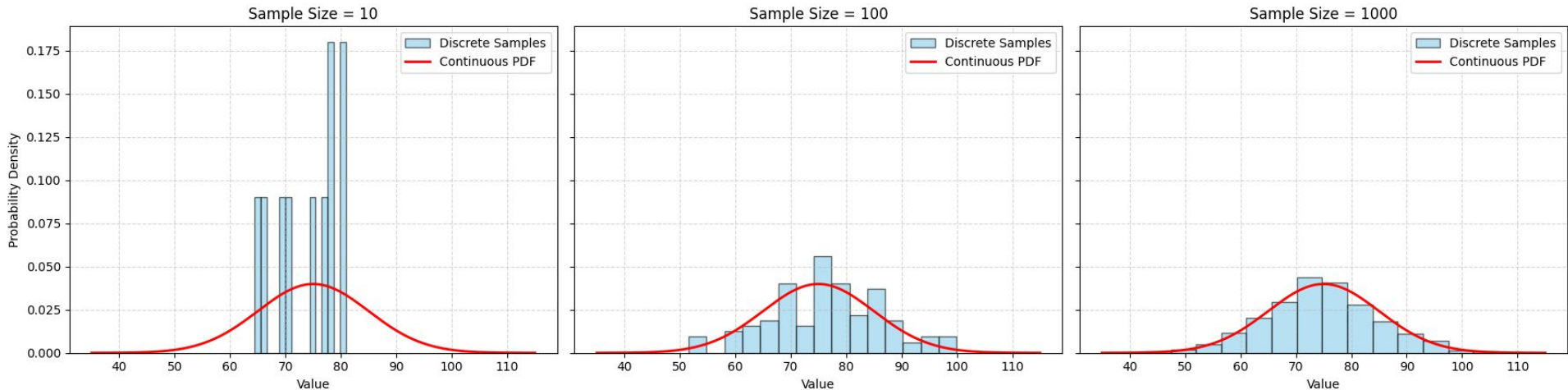
F jumps by $1/6$ at each of $1, \dots, 6$ and is flat in between (a step function)

$$\begin{aligned} \text{Ranges: } P(2 < X \leq 5) &= F(5) - F(2) \\ &= 5/6 - 2/6 = 1/2 \end{aligned}$$

Key idea: CDF $F(a) = P(X \leq a)$: add up the PMF over every value $\leq a$

This is just the discrete world!

Continuous PDF vs. Discrete Samples (Normal Distribution, Mean=75, SD=10)



Practice 6: Max of Two Dice

Roll two fair dice. Let M = the larger of the two values (if they tie, M is that value).

(a) Find the PMF of M : $P(M = k)$ for $k = 1, \dots, 6$.

(b) Check that your PMF sums to 1.

(c) Find $P(M \leq 3)$ and $P(M \geq 5)$.

Practice 6: Solution

(a) $M \leq k$ means both dice $\leq k$: k^2 of the 36 outcomes

$$P(M = k) = P(M \leq k) - P(M \leq k - 1) = (k^2 - (k - 1)^2) / 36, \text{ so}$$

$$P(M = k) = \begin{cases} (2k - 1) / 36 & \text{if } k \in \{1, 2, \dots, 6\} \\ 0 & \text{otherwise} \end{cases}$$

$$k = 1, 2, \dots, 6: \quad 1/36, 3/36, 5/36, 7/36, 9/36, 11/36$$

(b) $1 + 3 + 5 + 7 + 9 + 11 = 36 \rightarrow 36/36 = 1 \checkmark$

(c) $P(M \leq 3) = 9/36 = 1/4$; $P(M \geq 5) = (9 + 11)/36 = 20/36 = 5/9$

Answers: $P(M = k) = (2k - 1)/36$; $P(M \leq 3) = 1/4$; $P(M \geq 5) = 5/9$

The Binomial

This is super important

The Binomial Distribution

$X \sim \text{Bin}(n, p)$ counts successes in n independent trials,
each succeeding with the SAME probability p

PMF:

$$P(X = k) = \begin{cases} C(n, k) \cdot p^k \cdot (1 - p)^{n-k} & \text{if } k \in \{0, 1, \dots, n\} \\ 0 & \text{otherwise} \end{cases}$$

$C(n, k)$: which k trials are the successes

$p^k (1 - p)^{n-k}$: probability of one specific sequence with k successes

What is this?

Key idea: $X \sim \text{Bin}(n, p) \rightarrow P(X = k) = C(n, k) p^k (1 - p)^{n-k}$

Practice 7: Clinical Trial

A new medicine is tested on 40 patients. Each patient is cured independently with probability $3/10$.

Let X = the number of cured patients.

- (a) What distribution does X have?
- (b) What is $P(X = 38)$?
- (c) What is the most likely number of cures, and what is its probability?

Practice 7: Solution

(a) 40 independent trials, same $p = 0.3 \rightarrow X \sim \text{Bin}(40, 0.3)$

(b) $P(X = 38) = C(40, 38) (0.3)^{38} (0.7)^2 = 780 \cdot (0.3)^{38} \cdot 0.49 \approx 5.2 \times 10^{-18}$

Basically impossible: we expect around $40 \cdot 0.3 = 12$ cures

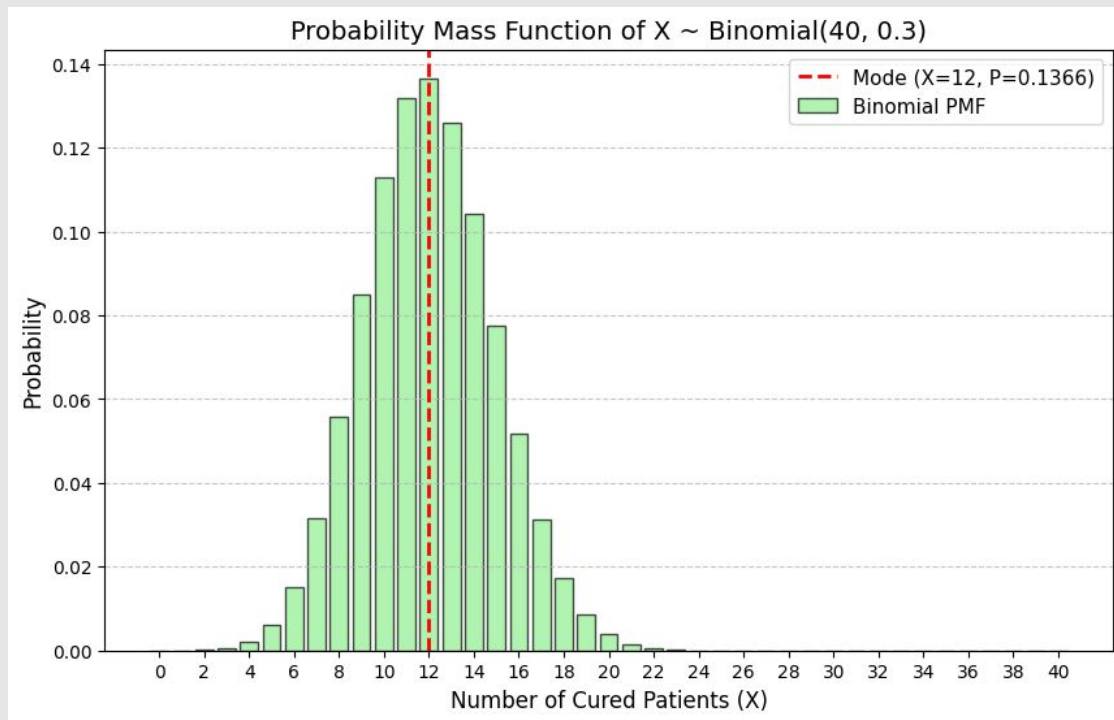
(c) The PMF peaks at $k = 12$:

$$P(X = 12) = C(40, 12) (0.3)^{12} (0.7)^{28} \approx 0.137$$

Even the most likely value has under a 14% chance (probability is spread out)

Answers: $X \sim \text{Bin}(40, 0.3)$; $P(X = 38) \approx 5.2 \times 10^{-18}$; mode 12, $P \approx 0.137$

Plot of `binom.pmf(np.arange(0,41), 40, .3)`



Practice 8: Planting Tomatoes

Each tomato seed sprouts independently with probability 0.7. You plant 5 seeds.

- (a) What is the probability that more than 3 tomatoes sprout?
- (b) A classmate answers (a) with $C(5, 4) \cdot (0.7)^4$. What's wrong with that?

Practice 8: Solution

(a) $X \sim \text{Bin}(5, 0.7)$. “More than 3” = $X = 4$ or $X = 5$ (mutually exclusive $\rightarrow$ add)

$$P(X = 4) = C(5, 4) (0.7)^4 (0.3)^1 = 5 \cdot 0.2401 \cdot 0.3 = 0.36015$$

$$P(X = 5) = (0.7)^5 = 0.16807$$

$$P(X > 3) = 0.36015 + 0.16807 = 0.52822$$

(b) $C(5, 4) (0.7)^4 = 1.2005 > 1$: impossible for a probability!

It ignores the (0.3) factor for the 5th seed, so a 5-sprout outcome gets counted 5 times

Answers: (a) $P(X > 3) = 0.52822$ (b) it double-counts (and exceeds 1)

The Bernoulli

What if we decompose a single binomial?

The Bernoulli Distribution

$X \sim \text{Ber}(p)$: ONE trial. $X = 1$ (success) with probability p , otherwise $X = 0$

PMF:

$$P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

What does this look like?

Key idea: $\text{Ber}(p)$ is one yes/no trial: $E[X] = p$, $\text{Var}(X) = p(1 - p)$

The Geometric Distribution

$X \sim \text{Geo}(p)$: repeat independent trials (each a success p) until the FIRST success
 X = the number of trials it took. Values: 1, 2, 3, ... (never 0)

PMF:

$$P(X = k) = \begin{cases} (1 - p)^{k-1} \cdot p & \text{if } k \in \{1, 2, 3, \dots\} \\ 0 & \text{otherwise} \end{cases}$$

Why??

Key idea: $\text{Geo}(p)$ waits for the first success: $P(X = k) = (1 - p)^{k-1} p$

Practice: Waiting for a 6

Roll a fair six-sided die until you get the first 6. Let X = the number of rolls.

- (a) What distribution does X have? Write its PMF.
- (b) What is $P(X = 3)$?
- (c) What is the probability that you need more than 4 rolls?
- (d) On average, how many rolls does it take?

Waiting for a 6: Solution

(a) Independent rolls, each a 6 w.p. $1/6$; stop at the first 6 $\rightarrow X \sim \text{Geo}(1/6)$

$$P(X = k) = \begin{cases} (5/6)^{k-1} \cdot (1/6) & \text{if } k \in \{1, 2, 3, \dots\} \\ 0 & \text{otherwise} \end{cases}$$

(b) $P(X = 3) = (5/6)^2 \cdot (1/6) = 25/216 \approx 0.116$ (two non-6s, then a 6)

(c) $P(X > 4) = (5/6)^4 = 625/1296 \approx 0.482$ (the first four rolls all miss)

(d) $E[X] = 1/p = 1/(1/6) = 6$ rolls

Answers: $\text{Geo}(1/6)$; $25/216 \approx 0.116$; $625/1296 \approx 0.482$; $E[X] = 6$

The Negative Binomial Distribution

$X \sim \text{NegBin}(r, p)$: repeat independent trials (success w.p. p) until the r -th success

X = the number of trials it took. Values: $r, r + 1, r + 2, \dots$

PMF:

$$P(X = k) = \begin{cases} C(k - 1, r - 1) \cdot p^r \cdot (1 - p)^{k-r} & \text{if } k \in \{r, r + 1, r + 2, \dots\} \\ 0 & \text{otherwise} \end{cases}$$

Can someone derive this?

Key idea: Bin: fixed n trials, count successes. NegBin: fixed r successes, count trials

NBA Finals: Problem

The NBA playoffs start with 16 teams.

Each series is best of 7 game (first to 4 wins)

The first round is 8 games, then 4, 2, 1

Assume every game is independent and the odds of winning each game is 50%

How many games do you expect there to be in the playoffs?

NBA Finals: Solution

There are $8 + 4 + 2 + 1 = 15$ playoff series, so we just need $E[N]$, the expected number of games in one best-of-7 series (each game an independent 50/50 outcome). N is 4, 5, 6, or 7:

$$P(N = 4) = 2 \cdot (1/2)^4 = 1/8$$

either team wins the first 4 games

$$P(N = 5) = 2 \cdot C(4, 3) \cdot (1/2)^5 = 1/4$$

winner is 3–1 after four games, then wins Game 5

$$P(N = 6) = 2 \cdot C(5, 3) \cdot (1/2)^6 = 5/16$$

winner is 3–2 after five games, then wins Game 6

$$P(N = 7) = C(6, 3) \cdot (1/2)^6 = 5/16$$

the teams must be tied 3–3 after six games

$$E[N] = 4 \cdot 1/8 + 5 \cdot 1/4 + 6 \cdot 5/16 + 7 \cdot 5/16 = 0.5 + 1.25 + 1.875 + 2.1875 = 5.8125 \text{ games per series}$$

$$15 \text{ series} \times 5.8125 = 87.1875 \text{ games} \quad (\text{range: 60 if every series is a sweep, 105 if all go to Game 7})$$

NegBin ($r = 4$, $p = 1/2$): game n is the winner's 4th win, so $P(N = n) = 2 \cdot C(n - 1, 3) \cdot (1/2)^n$

Answer: $15 \times 5.8125 = 87.1875$, so about 87.2 games in the playoffs

Practice 9 (Challenge): Random Teammates

12 people are split into teams with a random number generator: each person independently gets a team number from 1 to 6, uniformly at random.

What is the probability that at least one person has no teammate (ends up alone on a team)?

Practice 9: Solution (1/2)

Sample space: each of 12 people picks 1 of 6 teams $\rightarrow |S| = 6^{12}$ equally likely outcomes

Let A_i = “team i has exactly one member.” We want $P(A_1 \cup A_2 \cup \dots \cup A_6) \rightarrow$
inclusion–exclusion

Count outcomes where j specific teams are singletons:

choose the people for those teams, in order: $12 \cdot 11 \cdots (12 - j + 1) = 12! / (12 - j)!$

everyone else avoids those j teams: $(6 - j)^{(12 - j)}$

So for any j specific teams, $P(\text{all } j \text{ are singletons}) = [12! / (12 - j)!] \cdot (6 - j)^{(12 - j)} / 6^{12}$

Practice 9: Solution (2/2)

Inclusion–exclusion over the 6 teams:

$$P(\text{some singleton}) = \sum_{j=1}^6 (-1)^{j+1} C(6, j) \cdot P(j \text{ given teams are solo})$$

Equivalently: # outcomes with NO singleton team = 262,755,576

$$P = 1 - 262,755,576 / 2,176,782,336 \approx 1 - 0.121 = 0.879$$

Surprise: almost 88%! On average about 1.6 teams end up with a lone member
(Simulation check: 200,000 trials $\rightarrow$ 0.880)

Answer: $P \approx 0.879$ ($= 2,953,745 / 3,359,232$)

Someone ask me a question



Moments



Expectation: A Weighted Average

$E[X] = \sum x \cdot P(X = x)$ (loop over every value x , weighted by its probability)

Fair die: $E[X] = (1 + 2 + 3 + 4 + 5 + 6) / 6 = 3.5$

$E[X]$ doesn't have to be a value X can actually take

Classes of size 5, 10, 150. Pick a CLASS at random: $E[\text{size}] = 165 / 3 = 55$

Also called the mean, the average, or the first moment

Key idea: $E[X] = \sum x P(X = x)$, a probability-weighted average of the values

LOTUS: Expectation of a Function, $E[g(X)]$

$E[g(X)] = \sum g(x) \cdot P(X = x)$ (apply g to the VALUES; keep the probabilities)

Fair die: $E[X^2] = (1 + 4 + 9 + 16 + 25 + 36) / 6 = 91/6 \approx 15.17$

But $E[X]^2 = 3.5^2 = 12.25$, so $E[X^2] \neq (E[X])^2$ in general

Key idea: $E[g(X)] = \sum g(x) P(X = x)$. Transform the values, not the probabilities

Linearity of Expectation

$E[aX + b] = a E[X] + b$ Die winnings $W = 2X - 1$: $E[W] = 2(3.5) - 1 = 6$

$E[X + Y] = E[X] + E[Y]$ ALWAYS, even when X and Y are dependent

Sum of two dice: $E[X + Y] = 3.5 + 3.5 = 7$ (no 11-term sum needed)

Products are different: $E[XY] = E[X] E[Y]$ needs independence

$Y = 7 - X$: $E[XY] = 28/3 \approx 9.33$, but $E[X] E[Y] = 12.25$

Key idea: $E[\text{sum}] = \text{sum of the E's, always.}$

Indicator Variables

Indicator of event A: $I = 1$ if A happens, 0 if not (a Bernoulli(p) variable)

$$E[I] = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A)$$

Counting trick: write a count as a sum of indicators, then use linearity

$$X = I_1 + I_2 + I_3 + \dots \rightarrow E[X] = P(A_1) + P(A_2) + P(A_3) + \dots$$

Binomial: $X \sim \text{Bin}(n, p)$ is a sum of n Bernoulli(p) indicators $\rightarrow E[X] = np$

Key idea: $E[\text{indicator}] = P(\text{event})$, so $E[\text{count}] = \text{sum of the event probabilities}$

Worked Example: Matching Gifts

n people each bring a gift; the gifts are handed back in a random order

On average, how many people get their OWN gift back?

Direct way: $\sum k \cdot P(X = k)$ needs every $P(X = k)$, and even $P(X = 0)$ is hard

Indicator for person i : 1 if they get their own gift; its expectation is $1/n$

$X = \text{sum of the } n \text{ indicators} \rightarrow E[X] = n \cdot (1/n) = 1, \text{ for every } n$

The indicators are dependent (if $n - 1$ people match, so does the last), and that's fine

Key idea: sum the indicator probabilities: $E[\# \text{ matches}] = n \cdot (1/n) = 1$

Variance: How Spread Out Is X?

$\text{Var}(X) = E[(X - \mu)^2]$, where $\mu = E[X]$: the average squared distance from the mean

Computational formula: $\text{Var}(X) = E[X^2] - (E[X])^2$ (usually much easier)

Fair die: $\text{Var}(X) = 91/6 - 3.5^2 = 91/6 - 49/4 = 35/12 \approx 2.92$

Standard deviation: $\text{SD}(X) = \sqrt{\text{Var}(X)}$. Die: $\sqrt{35/12} \approx 1.71$

$\text{Var}(X) \geq 0$ always, and $\text{Var}(X) = 0$ only when X is a constant

Key idea: $\text{Var}(X) = E[X^2] - E[X]^2$. Mean = center; SD = typical distance from it

Var($aX + b$) and Standard Deviation

$$\text{Var}(aX + b) = a^2 \text{Var}(X) \quad \text{SD}(aX + b) = |a| \text{SD}(X)$$

Die winnings $W = 2X - 1$: $\text{Var}(W) = 4 \cdot 35/12 = 35/3 \approx 11.67$, $\text{SD}(W) \approx 3.42$

Trap: $\text{Var}(-X) = \text{Var}(X)$, not $-\text{Var}(X)$. Variance is never negative

$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ works for independent X, Y , but not in general

Moments of Classic Random Variables

Bernoulli $X \sim \text{Bern}(p)$: $E[X] = p$, $\text{Var}(X) = p(1 - p)$

Binomial $X \sim \text{Bin}(n, p)$: $E[X] = np$, $\text{Var}(X) = np(1 - p)$

n independent Bernoulli(p) trials: add the n means, and add the n variances

Geometric $X \sim \text{Geo}(p)$ (trials until the 1st success): $E[X] = 1/p$

Key idea: $\text{Bin}(n, p)$ has mean np and variance $np(1 - p)$.

Practice 10: Marble Points

Draw one marble: red with probability 0.5, green 0.3, blue 0.2.

Let $X = 1$ for red, 2 for green, 3 for blue.

- (a) Find $E[X]$.
- (b) Find $E[X^2]$ (use LOTUS).
- (c) Find $\text{Var}(X)$ and $\text{SD}(X)$.
- (d) You score $S = 10X - 4$ points. Find $E[S]$ and $\text{SD}(S)$.

Practice 10: Solution

(a) $E[X] = 1(0.5) + 2(0.3) + 3(0.2) = 0.5 + 0.6 + 0.6 = 1.7$

(b) $E[X^2] = 1(0.5) + 4(0.3) + 9(0.2) = 0.5 + 1.2 + 1.8 = 3.5$

(c) $\text{Var}(X) = E[X^2] - (E[X])^2 = 3.5 - 1.7^2 = 3.5 - 2.89 = 0.61$, $\text{SD}(X) = \sqrt{0.61} \approx 0.78$

(d) $E[S] = 10 E[X] - 4 = 17 - 4 = 13$

$\text{Var}(S) = 10^2 \cdot 0.61 = 61$, $\text{SD}(S) = 10 \cdot \text{SD}(X) = \sqrt{61} \approx 7.81$ (the -4 doesn't matter)

Answers: $E[X] = 1.7$, $E[X^2] = 3.5$, $\text{Var} = 0.61$, $\text{SD} \approx 0.78$; $E[S] = 13$, $\text{SD}(S) \approx 7.81$

Practice 11: Expected Max of Two Dice

Back to Practice 6: roll two fair dice and let M = the larger value, so $P(M = k) = (2k - 1)/36$.

(a) Find $E[M]$.

(b) Let L = the smaller value. Find $E[L]$ without working out a new PMF.

Hint for (b): what is $M + L$?

Practice 11: Solution

$$\begin{aligned} \text{(a) } E[M] &= \sum k \cdot (2k - 1)/36 = (1 \cdot 1 + 2 \cdot 3 + 3 \cdot 5 + 4 \cdot 7 + 5 \cdot 9 + 6 \cdot 11) / 36 \\ &= (1 + 6 + 15 + 28 + 45 + 66) / 36 = 161/36 \approx 4.47 \end{aligned}$$

(b) Every roll has $M + L = (\text{die 1}) + (\text{die 2})$, so by linearity:

$$E[M] + E[L] = E[\text{die 1} + \text{die 2}] = 7 \rightarrow E[L] = 7 - 161/36 = 91/36 \approx 2.53$$

M and L are dependent, but linearity doesn't care

$$\text{Check: } P(L = k) = (13 - 2k)/36 \text{ gives } (11 + 18 + 21 + 20 + 15 + 6)/36 = 91/36 \checkmark$$

Answers: $E[M] = 161/36 \approx 4.47$; $E[L] = 91/36 \approx 2.53$

Practice 12: Guessing on a Quiz

A quiz has 20 multiple-choice questions, 4 options each. You guess every answer at random.

Correct answer: +1 point. Wrong answer: $-1/3$ point. Let C = the number correct.

(a) What distribution does C have? Find $E[C]$ and $\text{Var}(C)$.

(b) Your score is $S = C - (20 - C)/3$. Find $E[S]$ and $\text{SD}(S)$.

(c) Why do you think the penalty is exactly $1/3$?

Practice 12: Solution

(a) 20 independent trials, $p = 1/4$: $C \sim \text{Bin}(20, 1/4)$, $E[C] = 5$, $\text{Var}(C) = 20 \cdot \frac{1}{4} \cdot \frac{3}{4} = 3.75$

(b) $S = C - (20 - C)/3 = (4/3)C - 20/3$ (a linear function of C)

$$E[S] = (4/3)(5) - 20/3 = 0$$

$$\text{Var}(S) = (4/3)^2 \cdot 3.75 = 20/3 \approx 6.67, \quad \text{SD}(S) \approx 2.58$$

(c) The penalty is chosen so that blind guessing is worth 0 points on average

Practice 13 (Challenge): Distinct Birthdays

30 people have independent birthdays, each uniform over 365 days (ignore leap years).

- (a) What is the expected number of DISTINCT birthdays among them?
- (b) What is the expected number of PAIRS of people who share a birthday?

Hint: use one indicator per day for (a) and one per pair of people for (b).

Practice 13: Solution

(a) For each day d : indicator = 1 if at least one person has birthday d

$$P(\text{day } d \text{ is used}) = 1 - (364/365)^{30} \approx 0.0790 \quad (\text{complement: nobody picks } d)$$

$$E[\# \text{ distinct}] = 365 \cdot (1 - (364/365)^{30}) \approx 28.84$$

(b) For each of the $C(30, 2) = 435$ pairs: indicator = 1 if they match

$$P(\text{a given pair matches}) = 1/365 \rightarrow E[\# \text{ matching pairs}] = 435/365 \approx 1.19$$

For Next Time

PSET 2: start early (due date on the course site)

Midterm Thu Oct 22, 6:30 pm. Start a running “formula + trap” sheet now

Take-home problems + solutions are at the end of these slides

Fill out the OH availability form if you haven't

This week's checkin form: <https://forms.gle/8uCGEidEzX7ZuBRQA>

More Practice (take-home)

Take-home 1: Phone Numbers

How many 7-digit phone numbers are possible if the first digit can't be 0 or 1?

Take-home 1: Solution

Product rule: 8 choices for the first digit (2–9), then 10 for each of the other 6

$$8 \cdot 10^6 = 8,000,000$$

Answer: 8,000,000

Take-home 2: Glitchy PSET App

Each time you press submit, the PSET app “glitches” and reveals the solution with probability 0.05, independently.

What is the probability of at least one glitch in 5 submissions?

Take-home 2: Solution

Let X = # glitches in 5 submissions, $X \sim \text{Bin}(5, 0.05)$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.95)^5 = 1 - 0.7738 \approx 0.226$$

Answer: $1 - 0.95^5 \approx 0.226$

Take-home 3: Chip Defect Detection

n chips are manufactured, and exactly 1 is defective. k chips are chosen at random for testing.

What is the probability that the defective chip is among the k tested chips?

Take-home 3: Solution

$|S| = C(n, k)$ equally likely sets of tested chips

$|E|$: the defective chip is in, plus $k - 1$ of the other $n - 1$ chips: $C(1, 1) \cdot C(n - 1, k - 1)$

$$P = C(n - 1, k - 1) / C(n, k) = k / n$$

Intuition: each chip has the same chance, k/n , of being tested

Answer: k / n

Take-home 4: More 1s Than 0s

A random 10-bit string has each bit independently equal to 1 with probability $1/2$.

What is the probability that it has more 1s than 0s?

Take-home 4: Solution

$X = \# \text{ of 1s} \sim \text{Bin}(10, 1/2)$. “More 1s than 0s” means $X \geq 6$

$$\begin{aligned} P(X \geq 6) &= [C(10,6) + C(10,7) + C(10,8) + C(10,9) + C(10,10)] / 2^{10} \\ &= (210 + 120 + 45 + 10 + 1) / 1024 = 386 / 1024 \approx 0.377 \end{aligned}$$

Symmetry check: $P(X \geq 6) = P(X \leq 4)$ and $P(X = 5) = 252/1024$, so $2 \cdot 386 + 252 = 1024$ ✓

Answer: $193/512 \approx 0.377$

Take-home 5: Combinatorial Proofs

Prove each identity by counting the same thing two ways (a story, not algebra):

(a) $C(n, k) = C(n, n - k)$

(b) $C(n, k) = C(n - 1, k - 1) + C(n - 1, k)$

(c) $C(n, 0) + C(n, 1) + \cdots + C(n, n) = 2^n$

Take-home 5: Solution

- (a) Choosing the k people who get cake = choosing the $n - k$ who don't
- (b) Fix one person, Pat. Groups of k either include Pat (choose $k - 1$ of the other $n - 1$) or don't (choose all k from the other $n - 1$). The cases don't overlap $\rightarrow$ add
- (c) Count subsets of n people: each person is in or out $\rightarrow 2^n$;
or sort the subsets by size $k \rightarrow$ sum of $C(n, k)$ over all k

Answer: Each side counts the same set of choices

Take-home 6: Round Table

9 people sit around a round table (rotations of the same seating count as the same).

How many seatings are there if 3 particular people must sit next to each other?

Take-home 6: Solution

Glue the 3 friends into one block $\rightarrow$ 7 “units” around the table

Circular arrangements of 7 units: fix one unit's seat, arrange the other 6 $\rightarrow 6! = 720$

Order the 3 friends inside the block: $3! = 6$

$$720 \cdot 6 = 4320$$

Answer: 4320

Take-home 7: Lines and Triangles

8 points lie in the plane, and exactly 3 of them are collinear (no other 3 are on a line).

(a) How many distinct straight lines pass through at least 2 of the points?

(b) How many triangles have their corners at these points?

Take-home 7: Solution

(a) Every pair gives a line: $C(8, 2) = 28$. But the 3 collinear points' $C(3, 2) = 3$ pairs all give the SAME line

$$28 - 3 + 1 = 26$$

(b) Every 3 points give a triangle, except the 3 collinear points: $C(8, 3) - 1 = 56 - 1 = 55$

Answers: (a) 26 (b) 55

Take-home 8: Seven Dice

Roll 7 fair six-sided dice. The probability that they sum to 10 can be written as $n / 6^7$. What is n ?

Take-home 8: Solution

Let $d_i = (\text{die } i) - 1 \geq 0$. Then $d_1 + \dots + d_7 = 10 - 7 = 3$

Stars and bars: 3 stars, 6 bars $\rightarrow C(3 + 6, 6) = C(9, 6) = 84$

(No d_i can exceed 5 since the total is only 3, so the upper limit never matters)

$$P = 84 / 6^7 = 84 / 279,936 \approx 0.0003$$

Answer: $n = 84$

Take-home 9: Entry Codes, Part 2

Entry codes are 6 digits and may start with 0.

- (a) How many codes contain a 5-digit palindrome (e.g., 123217 or 657775)?
- (b) How many codes contain exactly three of some digit in a row (e.g., 864443, 212221, 000777), with no run longer than 3?

Take-home 9: Solution

(a) Forms ABCBA? and ?ABCBA: 10^4 each $\rightarrow$ 20,000. Both at once: ABABAB (100)

Inclusion–exclusion: $20,000 - 100 = 19,900$

(b) Where is the run of 3 (D = repeated digit, $N \neq D$)?

DDD N ?? $\rightarrow 10 \cdot 9 \cdot 10 \cdot 10 = 9000$; N DDD N' ? $\rightarrow 10 \cdot 9 \cdot 9 \cdot 10 = 8100$

? N DDD N' $\rightarrow 8100$; ?? N DDD $\rightarrow 9000$

Overlap DDDddd (two runs) is counted twice: $10 \cdot 9 = 90$

$9000 + 8100 + 8100 + 9000 - 90 = 34,110$

Answers: (a) 19,900 (b) 34,110

Take-home 10: Double Straight

Six-card poker, continued: a “double straight” is 2 cards of one rank, 2 of the next rank, and 2 of the rank after that (e.g., 7 7 8 8 9 9).

Ranks run A, 2, ..., K, A (an ace can be low or high).

What is the probability of being dealt a double straight in a 6-card hand?

Take-home 10: Solution

$$|S| = C(52, 6) = 20,358,520$$

Starting rank: A, 2, ..., Q $\rightarrow$ 12 choices (A-2-3 up to Q-K-A); this fixes all three ranks

Suits: choose 2 of 4 for each rank $\rightarrow C(4, 2)^3 = 216$

$$P = 12 \cdot 216 / 20,358,520 = 2592 / 20,358,520 = 324 / 2,544,815 \approx 0.00013$$

(Every double straight is also a “three pair” hand)

Answer: $324 / 2,544,815 \approx 0.00013$